

ALGONQUIN COLLEGE

Faculty of Technology and Trades
School of Advanced Technology

Assignment 1

Digital Forensics Laboratory Proposal

in-house digital forensics laboratory

Name: Mohammad Zghoul

Student Number: 041097682

Course: CST8806 - Digital Forensics

Submitted to: Damien Bloor

Date of submission: June 06, 202

ALGONQUIN
COLLEGE

Spring 2026

Table of Contents

1. Overview	2
2. The Objective	2
3. The Actions	3
4. The Solution	3
5. Business Case	3
6. Our Proposal.....	4
6.1 Lab Location and Space Requirements	4
6.2 Physical Security and Environmental / Access Controls	4
6.3 Hardware Tools	4
Forensic Workstations	4
Write-Blockers	4
Jump Kits (Field Acquisition Kit)	5
Storage.....	5
6.4 Software Tools	5
Windows Forensic Tools	5
Linux Forensic Tools.....	6
6.5 Data Network	6
6.6 Lab Management and Case Management.....	6
7. Procedures, Policies, and Guidelines.....	6
Order of Volatility and Acquisition.....	6
Hashing and Verification.....	7
Write-Blocking Policy.....	7
Chain of Custody and Documentation.....	7
Antivirus, Malware, and Tool Validation.....	7
Legal and Professional Responsibilities (Canadian Context).....	7
8. Expected Results	7
Financial Benefits.....	7
Technical Benefits.....	7
Other Benefits	8
9. Pricing (Upfront and Operational Costs).....	8
10. Conclusion.....	9
11. References	10

1. Overview

This proposal suggests establishing an in-house digital forensics laboratory dedicated to our organization. Currently, all digital evidence acquisition, preservation, and analysis are outsourced to third-party forensic providers. This approach is expensive, slow, and requires us to relinquish custody of sensitive evidence to external parties—an increasingly hard position given the volume and confidentiality of our cases.

The proposed lab is envisioned as a comprehensive, single-source facility. It will address the four primary evidence types we encounter—computer/disk forensics, mobile-device forensics, network/log analysis, and volatile memory (RAM) forensics—covering everything from initial field response to courtroom-ready reports. This broad scope was chosen intentionally, rather than focusing on a niche discipline. Our caseload includes HR investigations, fraud, policy breaches, incident response, and litigation cases. A narrowly focused lab would merely shift our needs back to outsourcing when cases involve multiple disciplines. Defining this scope early ensures that physical layout, hardware, software, storage, networking, and procedures are aligned with the mission.

Fundamentally, the lab's design prioritizes evidentiary integrity. According to Section 31.1 of the Canada Evidence Act, the party seeking to admit an electronic document must prove its authenticity. Section 31.2 links the best evidence rule to verifying the integrity of the electronic record storage system (Canada Evidence Act, 1985). Every decision made in this proposal is evaluated by whether it helps establish, to a court's satisfaction, that the evidence was not altered by our actions.

2. The Objective

The purpose of this project is to stand up a forensically sound, secure, and legally defensible digital forensics laboratory that lets the organization investigate digital incidents internally - reducing cost and turnaround time while improving the confidentiality and integrity of evidence handling. The laboratory exists to serve the following needs:

- **Need #1 - Reduce the cost of third-party forensic services.** External per-case engagement and retainer fees, plus charging bench time, are consuming a growing share of the security budget. An internal capability converts that variable, escalating spend into a predictable fixed cost.
- **Need #2 - Preserve confidentiality and chain of custody.** Sending devices and disk images to outside vendors widens the circle of people who handle sensitive evidence and lengthens the chain of custody. An in-house lab keeps custody within the organization end-to-end.
- **Need #3 - Improve responsiveness and legal defensibility.** Outsourcing introduces queueing and shipping delays at the very moment volatile data (memory, running processes, network connections) is being lost. An internal team can respond immediately and follow a documented, repeatable methodology aligned with recognized standards (NIST SP 800-86; ISO/IEC 27037).

3. The Actions

To meet the needs above, the project is organized around three primary goals:

- **Goal #1 - Determine the scope of the digital forensics lab.** Confirm the evidence domains in scope (computer, mobile, network, and memory forensics), the expected caseload, and the standards the lab will be held to, so that every component can be sized against a clear mission.
- **Goal #2 - Select and validate the right hardware and software toolset.** Choose a balanced mix of commercial (COTS) and open-source tools, acquire the hardware that guarantees write-blocked acquisition, and validate each tool before it is used on live evidence.

Goal #3 - Establish a secure facility and a documented set of procedures. Build out a physically secure, access-controlled lab space and codify Standard Operating Procedures (SOPs) governing acquisition, hashing, storage, and chain of custody so results are repeatable and court-admissible.

4. The Solution

In summary, we propose to build a compact but complete forensic laboratory staffed by trained examiners and governed by formal procedure. The following recommendations lead to that solution:

- **Recommendation #1 - Build the lab in-house with a hybrid toolset.** Pair industry-standard commercial tools (Magnet AXIOM, Cellebrite, FTK, EnCase) with validated open-source tools (Autopsy, CAINE, SANS SIFT, Volatility) so that findings can be independently cross-verified and the budget is controlled without compromising scientific standards.
- **Recommendation #2 - Make evidence integrity non-negotiable through hardware write-blocking and hashing.** Acquire every source device through a hardware write-blocker (OpenText Tableau T8u and a multi-interface kit), verify every image with cryptographic hashing (MD5 plus SHA-256), and document the result, in keeping with section 31 of the Canada Evidence Act.
- **Recommendation #3 - Secure the facility and formalize the procedure.** House the lab in a restricted, access-logged room with a dedicated isolated network and evidence safe and operate it under written SOPs and a chain-of-custody regime, supported by ongoing examiner training and certification.

5. Business Case

The organization currently relies entirely on external forensic vendors. Beyond direct cost, this model carries three structural problems: it is slow (engagement, shipping, and queue time routinely add days to every matter); it dilutes confidentiality and lengthens the chain of custody by handing sensitive evidence to outside parties; and it leaves us dependent on third-party availability during exactly the incidents - active intrusions, suspected insider activity - where speed matters most.

An in-house laboratory addresses all three. The recurring annual operating cost of the proposed lab (approximately \$81,500 after the first year, as detailed in Section 9) is comparable to the cost of only a handful of complex outsourced engagements, yet it provides unlimited internal capacity, immediate response, and full custody control. Once caseload exceeds roughly a dozen significant matters per year, the lab pays for itself while simultaneously strengthening our legal position. The remainder of this proposal details how the laboratory will be designed, equipped, secured, and operated.

6. Our Proposal

This section details the proposed laboratory across its physical, technical, and management dimensions.

6.1 Lab Location and Space Requirements

The laboratory will occupy a dedicated, interior room of roughly 550-650 square feet, physically separated from general business operations so that examiners are not interrupted and evidence is never exposed to ordinary staff traffic. An interior location with no exterior windows reduces both the risk of surveillance and the risk of forced entry. The room will be partitioned into three functional zones: an evidence intake and storage zone (containing the safe evidence and logging station), an examination zone (workstations and write-blocking hardware), and a small report-writing/administrative zone. Bench space and power will be sized to allow two examiners to work concurrently on separate cases without cross-contamination.

6.2 Physical Security and Environmental / Access Controls

Because the integrity of the environment is central to the admissibility of evidence, physical security is treated as a first-class requirement rather than an afterthought:

- **Access control:** the lab door is secured by badge/PIN access with an automatically generated entry log; access is restricted to named forensic personnel on a least privilege basis.
- **Monitoring:** CCTV covering the entrance, evidence safe, and examination benches, with retained footage, provides an independent record of who handled what and when.
- **Evidence storage:** fire-rated, lockable evidence safe stores original media; the safe is itself inside the access-controlled room, providing layered (defence-in-depth) protection.
- **Environmental controls:** dedicated climate control and an uninterruptible power supply (UPS) protect equipment and prevent data loss from power events; Faraday/shielded storage isolates mobile devices from networks to prevent remote wiping.
- **Chain-of-custody enforcement:** every item entering or leaving the safe is recorded in a custody log, so that custody can be demonstrated continuously from seizure to court.

6.3 Hardware Tools

The hardware program is built around one principle: original evidence must never be altered during acquisition or examination.

Forensic Workstations

Two high-specification forensic workstations form the analytical core. Each is configured with a multi-core CPU, 128 GB of RAM (memory-image analysis and modern forensic suites are memory-hungry), fast NVMe scratch storage for processing, and large internal storage for working copies. High RAM and CPU directly reduce processing time on large images, which is one of the bottlenecks that currently makes outsourcing attractive. A separate mobile/triage laptop running a forensic Linux distribution supports on-scene acquisition.

Write-Blockers

A hardware write-blocker is placed between the forensic workstation and any source device. It physically intercepts and prevents write commands from reaching the source, so the act of imaging cannot modify the original - the technical guarantee that underpins section 31 of the Canada Evidence Act (Canada Evidence Act, 1985). We recommend the OpenText (formerly Guidance Software) Tableau Forensic USB 3.0 Bridge (T8u) for USB media, complemented by a multi-interface kit covering SATA, IDE, and PCIe/NVMe sources. The T8u has been independently tested under the NIST Computer Forensics Tool Testing program and sustains imaging

throughput more than 300 MB/s while computing MD5 and SHA-1 hashes during acquisition (OpenText, n.d.). Where a hardware blocker is unavailable, the SOP requires software write-blocking via a forensic boot environment (mounting read-only under Linux, or registry-based USB write protection under Windows) as a documented fallback.

Jump Kits (Field Acquisition Kit)

A pre-assembled jump kit allows examiners to respond on-site and capture evidence before it is lost or contaminated. The kit contains:

- A ruggedized field laptop with a bootable forensic OS (CAINE / SANS SIFT) for live acquisition;
- A portable hardware write-blocker plus a full set of drive adapters and cables (SATA, IDE, M.2, USB, SAS);
- Sterile (forensically wiped) target drives, write-once where possible, for capturing images;
- Faraday bags and a shielded pouch to isolate seized mobile devices from cellular and Wi-Fi networks, preventing remote wipe;
- Evidence bags, tamper-evident seals, labels, a camera, and chain-of-custody forms for documentation;
- Portable power, a USB memory-capture toolkit, and a hardware hashing/imaging device for rapid duplication.

Storage

Storage is separated into three tiers to protect integrity and support case management. (1) A sterile media pool of forensically wiped target drives is used to receive new images. (2) Working storage on a RAID-protected network-attached storage (NAS) array (roughly 80 TB usable) holds active case images and is sized for several large cases concurrently. (3) Archival/backup storage retains closed-case images for the required retention period, with a second copy held off the primary array. All evidence storage is logically and physically separate from corporate IT storage.

Solid-state drives raise a specific forensic concern the lab must plan for. On SSDs, the drive's garbage-collection process relocates valid data and erases old blocks autonomously, so unallocated space is frequently wiped and rarely recoverable, and - critically - re-imaging the same SSD can yield a different hash value even when no examiner action caused the change. The SOP therefore requires examiners to record this behaviour, image SSDs as promptly as possible, and document any hash variance with an explanation, so that an honest, well-understood limitation is never mistaken for evidence tampering.

6.4 Software Tools

The laboratory adopts a hybrid software strategy, combining commercial (COTS) suites with validated open-source tools. Commercial tools provide vendor support, regular updates, and broad device coverage; open-source tools are transparent, auditable, and free, and they allow findings to be independently cross-verified using a second, unrelated tool - a powerful way to demonstrate technical accuracy in court. The assignment's reference material rightly cautions that cost savings from open-source software must never come at the expense of scientific standards or the Canada Evidence Act; accordingly, every tool - commercial or open-source - is validated against known data before being used on live evidence, and open-source tools are assessed for forensic accuracy, process integrity, peer review, and community vetting.

Windows Forensic Tools

- **Magnet AXIOM** - a unified computer, mobile, and cloud platform; strong artifact parsing (browser history, chat, cloud) and reporting, used as the primary analysis suite.

- **Exterro FTK (Forensic Toolkit)** - indexing-driven analysis and the widely used free FTK Imager for forensically sound acquisition and verified imaging.
- **OpenText EnCase Forensic** - a long-established, court-recognized examination platform used for deep disk analysis and as a cross-verification tool.
- **Volatility (memory) and Cellebrite UFED (mobile)** - Volatility for RAM analysis (processes, network connections, injected code) and Cellebrite for mobile extraction and decoding, all running on or alongside the Windows workstations.

Linux Forensic Tools

- **CAINE (Computer Aided INvestigative Environment)** - a bootable Linux distribution that mounts evidence read-only by default, ideal for field acquisition and write-blocked imaging.
- **SANS SIFT Workstation** - a free, well-maintained Ubuntu-based analysis platform bundling a large open-source toolset for disk, memory, and timeline analysis.
- **Autopsy / The Sleuth Kit** - an open-source examination GUI and underlying toolkit for file-system analysis, keyword search, and reporting; used to independently corroborate commercial-tool findings.
- **Paladin and command-line utilities** - Paladin for imaging plus core utilities (dd/dcfldd, ewfacquire, and the hashing tools) for scripted, auditable acquisition and verification.

6.5 Data Network

The laboratory operates on a dedicated, isolated (air-gapped) network that is physically and logically separate from the corporate LAN and is not, by default, connected to the internet. This prevents malware on examined evidence from reaching production systems and prevents accidental contamination of evidence. A controlled, monitored connection - used only for licence activation, definition updates, and approved cloud-evidence acquisition - is available on an as-needed basis and is logged. Examiner accounts are individual (no shared logins) so that all activity is attributable.

6.6 Lab Management and Case Management

A case-management system records each case from intake to disposition: a unique case number, the custodian, every item of evidence, its hash values, who accessed it, and when. This system is the digital backbone of chain of custody and directly supports the integrity requirement of section 31.2 of the Canada Evidence Act. Management responsibilities include enforcing the SOPs, scheduling tool validation and proficiency testing, controlling access, and overseeing examiner training and certification (for example GCFA, EnCE, and vendor-specific courses) so the team's competence keeps pace with evolving technology.

7. Procedures, Policies, and Guidelines

Tools and facilities are only as defensible as the procedures that govern their use. The laboratory will operate under a documented set of Standard Operating Procedures (SOPs), aligned with NIST SP 800-86 and ISO/IEC 27037, covering at minimum the following.

Order of Volatility and Acquisition

When a system is live, evidence is captured from most volatile to least volatile so that fragile data is not lost: (1) memory (RAM), (2) running processes, (3) network connections and activity, (4) system settings, and finally (5) physical/disk acquisition (Kent et al., 2006). Acquisition prefers a full physical image over a logical copy where lawful and feasible, and the SOP addresses the encryption-versus-logical trade-off when full-disk encryption prevents physical capture.

Hashing and Verification

Immediately after acquisition, the lab computes a cryptographic hash (MD5 together with SHA-256) of both the source and the image and records the values. Matching hashes prove the image is a bit-for-bit duplicate of the original; the hash is recomputed and re-verified before each examination to demonstrate the evidence has not changed in storage. This is the practical mechanism by which the lab meets its burden under section 31.1/31.2 of the Canada Evidence Act.

Write-Blocking Policy

All acquisition is performed through a hardware write-blocker. Software write-blocking (read-only mounting under Linux, registry-based USB write protection under Windows) is permitted only as a documented fallback and, per best practice, only in combination with a forensic acquisition operating system (CAINE, SANS SIFT, or Paladin).

Chain of Custody and Documentation

Every transfer of evidence is logged with date, time, and the individuals involved, from seizure through analysis to courtroom presentation. Original media is never examined directly; examiners work only on verified working copies. Where appropriate, examiners apply digital signatures to reports and evidence records so that the authorship and integrity of the documentation itself can be proven.

Antivirus, Malware, and Tool Validation

Evidence is presumed potentially malicious and is handled on the isolated network; antivirus/anti-malware is used to flag (not silently quarantine) hostile content, so its presence is recorded rather than destroyed. Every tool is validated against known reference data before first use and re-validated after major updates, and examiners participate in periodic proficiency testing.

Legal and Professional Responsibilities (Canadian Context)

As Canadian security professionals, examiners are bound by the Canada Evidence Act and must exercise due diligence to ensure that not one bit of data is inadvertently modified by their actions. The SOPs therefore embed the Act's authenticity (s. 31.1) and integrity/best-evidence (s. 31.2-31.3) requirements directly into daily practice, and examiners always operate within lawful authority and organizational policy.

8. Expected Results

The approach above is expected to deliver measurable benefits because each design choice maps directly to one of the organization's stated needs.

Financial Benefits

- **Result #1:** Elimination of recurring per-case third-party fees - converting unpredictable variable spend into a fixed, plannable operating cost.
- **Result #2:** A favourable break-even point - the lab's annual operating cost is recovered once internal caseload exceeds roughly a dozen complex matters per year.
- **Result #3:** Reduced legal exposure and downstream cost - stronger, well-documented evidence handling lowers the risk of evidence being excluded or challenged.

Technical Benefits

- **Result #1:** Faster turnaround - immediate, in-house response captures volatile data before it is lost and removes shipping/queue delays.

- **Result #2:** Higher confidence in findings - a hybrid commercial/open-source toolset allows independent cross-verification of every key result.
- **Result #3:** Court-ready, repeatable output - standardized SOPs, hashing, and write-blocking produce results that are defensible and reproducible.

Other Benefits

Less tangible but real benefits include greater confidentiality (sensitive evidence never leaves the organization), improved morale and capability within the security team as examiners develop specialized, certified skills, and increased stakeholder and client confidence that incidents are handled rigorously and lawfully in-house.

9. Pricing (Upfront and Operational Costs)

The table below details estimated costs for establishing and operating the laboratory, grouped into upfront capital hardware, annual software licensing, and annual services/operational costs. Figures are realistic market estimates in Canadian dollars and will be confirmed by formal quotation during procurement; some commercial forensic licences are quoted only on request, so those values are best-available estimates.

Item	Cost (CAD)
Category 1 - Capital / Upfront Hardware (one-time)	
Forensic acquisition & analysis workstations (2 x \$9,500)	\$19,000
Mobile / triage forensic laptop (1)	\$4,500
Hardware write-blockers - OpenText Tableau T8u USB + SATA/IDE/PCIe kit (2 sets)	\$3,000
Field jump kit (Faraday bags, adapters, cables, imager, power)	\$5,500
Network-attached evidence storage (NAS, ~80 TB usable, RAID)	\$12,000
Backup / archival storage & enterprise HDDs/SSDs (sterile media pool)	\$6,000
Evidence safe, lockable racks, KVM, UPS, shredder	\$5,500
Total Category 1 (Upfront Hardware)	\$55,500
Category 2 - Software Licensing (annual)	
Magnet AXIOM (1 seat: computer + mobile + cloud)	\$5,500/yr
Cellebrite UFED + Physical Analyzer (mobile)	\$12,000/yr
Exterro FTK (1 seat)	\$4,000/yr
OpenText EnCase Forensic (1 seat)	\$3,500/yr
X-Ways Forensics (1 licence + maintenance)	\$1,600/yr
Open-source suite (Autopsy, CAINE, SANS SIFT, Volatility)	\$0
Endpoint protection, OS & productivity licences	\$1,400/yr
Total Category 2 (Annual Software)	\$28,000/yr
Category 3 - Services & Operational (annual)	
Facility lease & utilities (secure ~600 sq ft suite)	\$30,000/yr
Training & certifications (GCFA, EnCE, vendor courses)	\$12,000/yr

Item	Cost (CAD)
Accreditation, validation & proficiency testing (ISO/IEC 17025 alignment)	\$6,000/yr
Maintenance, insurance, shipping/handling, travel	\$5,500/yr
Total Category 3 (Annual Services)	\$53,500/yr
TOTAL - Year 1 (upfront + annual operating)	\$137,000
Recurring annual operating cost (Year 2+)	\$81,500/yr

Note: Hardware is a one-time capital cost; software and services recur annually. The Year-1 total includes upfront hardware plus the first year of operating cost. Even at this scale, the recurring cost is modest relative to sustained third-party outsourcing.

10. Conclusion

Digital forensics is a specialized, demanding discipline in which the integrity of the environment - the lab, the tools, and the procedures - is inseparable from the value of the evidence it produces. Continuing to outsource that work leaves the organization slower to respond, exposed on confidentiality, and dependent on others to safeguard evidence whose admissibility ultimately rests on our own due diligence. The laboratory proposed here resolves those problems: it is scoped as a full-service capability, equipped with validated commercial and open-source tools, anchored by hardware write-blocking and cryptographic hashing, secured physically and on an isolated network, and governed by SOPs that embed the Canada Evidence Act's authenticity and integrity requirements into everyday practice. The result is faster, cheaper, more confidential, and more legally defensible forensic work. Establishing this laboratory is a sound investment in the organization's ability to investigate digital incidents thoroughly, lawfully, and with evidence that will withstand scrutiny in court.

11. References

- Canada Evidence Act, RSC 1985, c. C-5, ss. 31.1-31.3. Justice Laws Website. <https://laws-lois.justice.gc.ca/eng/acts/C-5/section-31.1.html>
- Kent, K., Chevalier, S., Grance, T., & Dang, H. (2006). *Guide to integrating forensic techniques into incident response (NIST Special Publication 800-86)*. National Institute of Standards and Technology. <https://nvlpubs.nist.gov/nistpubs/legacy/sp/nistspecialpublication800-86.pdf>
- International Organization for Standardization. (2012). *Information technology - Security techniques - Guidelines for identification, collection, acquisition and preservation of digital evidence (ISO/IEC 27037:2012)*. ISO.
- OpenText. (n.d.). *Tableau Forensic USB 3.0 Bridge (T8u) - product brief*. OpenText Security. <https://security.opentext.com/tableau/hardware/details/t8u>
- Scientific Working Group on Digital Evidence. (2018). *SWGDE best practices for computer forensic acquisitions*. SWGDE. <https://www.swgde.org>
- Carrier, B. (n.d.). *Autopsy & The Sleuth Kit - open source digital forensics*. Basis Technology. <https://www.autopsy.com>
- SANS Institute. (n.d.). *SIFT Workstation*. SANS. <https://www.sans.org/tools/sift-workstation/>